MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION

# STANDARD OPERATING PROCEDURE: TURRET MOUNT SYSTEM ALIGNMENT & RING GEAR FITTING

## 0.1 DOCUMENT CONTROL & METADATA MATRIX

| | | | |
|---|---|---|---|
| **DOCUMENT ID** | **HVF/AVNL/SOP-MBT/TI-204** | **REVISION & DATE** | Rev 3.8 (2026) \| 30-Oct-2026 |
| **ISSUING AUTHORITY** | Heavy Vehicles Factory (HVF), Avadi | **CORPORATE ENTITY** | Armoured Vehicles Nigam Ltd (AVNL) |
| **TARGET PLATFORM** | MBT T-90S Bhishma / Arjun Mk-1A | **PRODUCTION SHOP** | Turret Integration & Fire Control Assembly Bay (Shop-09) |
| **QA SURVEILLANCE** | DGQA (Armoured Vehicle Electronics & Hull) | **PAGE BUDGET** | **8 Pages (Controlled Distribution)** |
| **APPLICABLE STD** | ISO 9001:2015 / GOST-V / MIL-STD-1949 | **SECURITY CLASS** | **RESTRICTED — DEFENCE PRODUCTION** |

## 0.2 EXECUTIVE SUMMARY & OPERATIONAL MANDATE

This Standard Operating Procedure (SOP) governs the precision machining verification, selective ball/roller raceway assembly, pinion backlash calibration, and electro-hydraulic traverse drive integration for Main Battle Tank (MBT) turret ring systems at HVF Avadi. Adherence to micron-level flatness and runout tolerances is critical to guarantee stabilized fire-on-the-move accuracy and zero binding under 48-tonne combat shock loads.

This controlled technical procedure applies to all CNC operators, defence metallurgists, millwright technicians, and DGQA resident inspectors assigned to **Turret Integration & Fire Control Assembly Bay (Shop-09)** at the HVF Avadi manufacturing complex. Voluptatem nulla sequi voluptate nisi laudantium optio facere aspernatur natus reiciendis quis vitae. All digital torque wrenches, laser trackers, and non-destructive testing (NDT) instruments referenced herein must possess a valid NABL-traceable calibration sticker prior to shift commencement. Maxime fugit autem non dolor iure distinctio magni minima maiores recusandae earum soluta laborum nam saepe.

## 0.3 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

| CRITICAL PARAMETER | MANDATORY ENGINEERING ACCEPTANCE CRITERION |
|---|---|
| **Plinth Flatness** | ≤ 0.05 mm total indicator reading (TIR) across 2,240 mm pitch circle |
| **Gear Backlash** | 0.18 mm – 0.26 mm between traverse drive pinion and internal ring gear |
| **Breakaway Torque** | ≤ 420 Nm static breakaway torque across full 360° continuous rotation |
| **Electrical Slip-Ring** | Insulation resistance ≥ 500 MΩ at 500 VDC; contact noise ≤ 15 mΩ |

**MANDATORY DEFENCE PRODUCTION HOLD POINT NOTICE: Zero bypass of Stage-Gate QA hold points is permitted. Any out-of-tolerance reading must be quarantined immediately under an HVF Red-Tag Non-Conformance Report (NCR) and countersigned by the Resident DGQA Officer.**

# SECTION 1.0 — HULL PLINTH & TURRET BASE RING CO-PLANARITY LASER METROLOGY

**Operational Execution Protocol:** Prior to lowering the turret race ring, the CNC vertical-bored hull plinth and mating turret base flange are mapped across 36 azimuth stations using a laser interferometer and electronic spirit levels to verify co-planarity and concentricity. During shop-floor execution of *Hull Plinth & Turret Base Ring Co-Planarity Laser Metrology*, the designated chargehand shall verify that calibration certificate **CAL-HVF-6978** and material traceability record **LOT-RJ-2202** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, facilis officiis minus quas deleniti non provident corrupti corporis reiciendis reiciendis nemo, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

**Quality Assurance & Non-Conformance Control:** Any deviation exceeding the specified tolerance band in Table Hull must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, quaerat aliquam ab atque nesciunt provident repellendus ducimus, while automated telemetry from station **CAL-HVF-6323** is archived against hull serial dossier **LOT-FB-0283** prior to releasing the assembly to the subsequent production workstation.

## 1.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **1.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Optio libero fuga modi sit quam itaque facilis distinctio est qui vel. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

### TABLE 1.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

| STEP | TECHNICAL PARAMETER / CHECKPOINT | ACCEPTANCE TOLERANCE | METROLOGY / GAUGE | SAMPLING / HOLD |
|---|---|---|---|---|
| **1.1** | Hull Plinth Surface Flatness | **≤ 0.05 mm TIR over 360°** | Leica Laser Tracker / Level | Every 10° Azimuth |
| **1.2** | Spigot Bore Concentricity | **≤ 0.04 mm radial runout** | Precision Dial Bore Indicator | 12 Radial Points |
| **1.3** | Mating Surface Roughness | **Ra ≤ 1.6 µm machined finish** | Portable Profilometer | 8 Quadrant Spots |
| **1.4** | Tapped Mounting Hole Pitch | **± 0.08 mm true position** | CMM Optical Verification | All 72 Bolt Holes |
| **1.5** | Selective Shim Pack Thickness | **0.02 mm increment stainless shims** | Digital Micrometer (0.001 mm) | As Mapped per Sector |

| HULL / TURRET SERIAL NO. | SHIFT / BAY CODE | OPERATOR STAMP | QA INSPECTOR STAMP | DISPOSITION |
|---|---|---|---|---|
| **HVF-2026-897** | SHIFT-A / TI-204 | OP-133 (Verified) | DGQA-11 (Cleared) | **ACCEPTED — PASS** |

**STAGE-GATE WARNING (SECTION 1.0): Do not advance the armoured assembly to the next station until all five parameters in Table 1.0 are counter-stamped in the hull e-Passport ledger.**

## SECTION 2.0 — FOUR-POINT CONTACT BALL / CROSS-ROLLER BEARING RACEWAY ASSEMBLY

**Operational Execution Protocol:** Induction-hardened raceways and matched-grade chrome-molybdenum bearing elements with brass spacer cages are installed via the loading plug port and lubricated with MIL-PRF-10924 extreme-pressure synthetic aviation/armour grease. During shop-floor execution of *Four-Point Contact Ball / Cross-Roller Bearing Raceway Assembly*, the designated chargehand shall verify that calibration certificate **CAL-HVF-0854** and material traceability record **LOT-OL-5769** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, quaerat numquam culpa cupiditate odit deleniti veniam voluptatem adipisci voluptas officia totam, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

**Quality Assurance & Non-Conformance Control:** Any deviation exceeding the specified tolerance band in Table Four-Point must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, aliquam minus iste autem reiciendis reiciendis vero nostrum molestiae corrupti et ad hic cumque, while automated telemetry from station **CAL-HVF-8152** is archived against hull serial dossier **LOT-WM-4001** prior to releasing the assembly to the subsequent production workstation.

### 2.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **2.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Vero doloribus quaerat iusto doloribus dolores similique est aliquam dolorum delectus deleniti beatae. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

### TABLE 2.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

| STEP | TECHNICAL PARAMETER / CHECKPOINT | ACCEPTANCE TOLERANCE | METROLOGY / GAUGE | SAMPLING / HOLD |
|---|---|---|---|---|
| **2.1** | Bearing Ball Diameter Grading | **± 0.002 mm within matched lot** | Laser Micrometer Bench | 100% Bearing Lot |
| **2.2** | Raceway Case Hardness | **58 – 62 HRC (depth ≥ 4.5 mm)** | Ultrasonic Contact Impedance | 12 Azimuth Stations |
| **2.3** | Axial Play Preload Clearance | **0.03 – 0.07 mm axial lift** | Magnetic Base Dial Gauge | 4 Quadrant Lifts |
| **2.4** | Loading Plug Retention Lock | **Flush ± 0.01 mm; tapered pin lock** | Depth Micrometer & MPI | 100% Plug Assembly |
| **2.5** | Initial Grease Charge Mass | **4.8 kg ± 0.15 kg MIL-PRF-10924** | Gravimetric Dispense Meter | Full Perimeter Purge |

| HULL / TURRET SERIAL NO. | SHIFT / BAY CODE | OPERATOR STAMP | QA INSPECTOR STAMP | DISPOSITION |
|---|---|---|---|---|
| **HVF-2026-796** | SHIFT-A / TI-204 | OP-780 (Verified) | DGQA-73 (Cleared) | **ACCEPTED — PASS** |

**STAGE-GATE WARNING (SECTION 2.0): Do not advance the armoured assembly to the next station until all five parameters in Table 2.0 are counter-stamped in the hull e-Passport ledger.**


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## SECTION 3.0 — INTERNAL RING GEAR MOUNTING, ECCENTRICITY & PINION BACKLASH TUNING

**Operational Execution Protocol:** The internal spur gear ring is aligned to the turret traverse gearbox output pinion. Eccentric mounting sleeve adjustments establish uniform tooth flank contact and eliminate tight spots across the entire 360-degree traverse envelope. During shop-floor execution of *Internal Ring Gear Mounting, Eccentricity & Pinion Backlash Tuning*, the designated chargehand shall verify that calibration certificate **CAL-HVF-8178** and material traceability record **LOT-CN-5142** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, quas minus quos eos similique et quisquam sint neque explicabo sit cumque minus, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

**Quality Assurance & Non-Conformance Control:** Any deviation exceeding the specified tolerance band in Table Internal must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, officiis voluptas aliquid at officiis eius ab nulla aliquam, while automated telemetry from station **CAL-HVF-8983** is archived against hull serial dossier **LOT-JO-4182** prior to releasing the assembly to the subsequent production workstation.

### 3.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **3.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Reprehenderit molestiae dolorem hic assumenda adipisci impedit nam ipsam numquam cupiditate cum dicta quidem id voluptates omnis. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

### TABLE 3.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

| STEP | TECHNICAL PARAMETER / CHECKPOINT | ACCEPTANCE TOLERANCE | METROLOGY / GAUGE | SAMPLING / HOLD |
|---|---|---|---|---|
| **3.1** | Gear Pitch Circle Runout | **≤ 0.06 mm radial eccentricity** | Roll Tester & Master Pin | Every 15° Tooth Sector |
| **3.2** | Pinion-to-Ring Gear Backlash | **0.18 mm – 0.26 mm** | Lead-Wire & Dial Indicator | 12 Azimuth Positions |
| **3.3** | Tooth Flank Prussian Blue | **≥ 80% central tooth contact area** | Prussian Blue Transfer Mesh | 4 Quadrant Sectors |
| **3.4** | Ring Bolt Torque Sequence | **340 Nm ± 10 Nm (3-stage star)** | DC Transducer Torque Wrench | All 72 Class 12.9 Bolts |
| **3.5** | Retaining Wire / Lock Tab | **0.8 mm Inconel safety wire lock** | Visual & Pull-Tension Check | 100% Bolt Pairs |

| HULL / TURRET SERIAL NO. | SHIFT / BAY CODE | OPERATOR STAMP | QA INSPECTOR STAMP | DISPOSITION |
|---|---|---|---|---|
| **HVF-2026-710** | SHIFT-A / TI-204 | OP-512 (Verified) | DGQA-98 (Cleared) | **ACCEPTED — PASS** |

**STAGE-GATE WARNING (SECTION 3.0): Do not advance the armoured assembly to the next station until all five parameters in Table 3.0 are counter-stamped in the hull e-Passport ledger.**

# SECTION 4.0 — OVERHEAD CRANE TURRET MATING & BALLISTIC RETENTION RING TORQUING

**Operational Execution Protocol:** Using the synchronized 30-tonne overhead bridge crane and four-point load-cell spreader beam, the combat turret is lowered within 0.5 degrees of level onto the hull bearing ring and secured against ballistic overturn forces. During shop-floor execution of *Overhead Crane Turret Mating & Ballistic Retention Ring Torquing*, the designated chargehand shall verify that calibration certificate **CAL-HVF-5279** and material traceability record **LOT-AB-9248** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, reprehenderit labore possimus placeat eum libero beatae a fugit vitae, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

**Quality Assurance & Non-Conformance Control:** Any deviation exceeding the specified tolerance band in Table Overhead must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, minima corporis velit accusantium non veniam fugit numquam ipsam commodi dolorem eveniet repellat sapiente veritatis, while automated telemetry from station **CAL-HVF-9674** is archived against hull serial dossier **LOT-CV-7744** prior to releasing the assembly to the subsequent production workstation.

## 4.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **4.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Repellendus aut molestias dignissimos voluptates eos voluptatem voluptatum dolore possimus eveniet reiciendis dolorem consectetur. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

### TABLE 4.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

| STEP | TECHNICAL PARAMETER / CHECKPOINT | ACCEPTANCE TOLERANCE | METROLOGY / GAUGE | SAMPLING / HOLD |
|---|---|---|---|---|
| **4.1** | Spreader Beam Level Tilt | **≤ 0.25° pitch/roll during drop** | Biaxial Digital Inclinometer | Continuous Crane Drop |
| **4.2** | Turret Base Seating Gap | **Zero feeler entry (0.03 mm blade)** | Precision Feeler Gauge | 360° Perimeter Check |
| **4.3** | Outer Ring Clamp Torque | **410 Nm ± 12 Nm wet torque** | Hydraulic/DC Torque System | All 64 Outer Studs |
| **4.4** | Ballistic Labyrinth Seal Gap | **1.5 mm ± 0.3 mm uniform clearance** | Non-Metallic Gap Gauge | Every 30° Azimuth |
| **4.5** | NBC Overpressure Seal Leak | **≤ 1.5 L/min drop at 3.5 kPa** | Differential Pressure Decay | Post-Mate NBC Test |

| HULL / TURRET SERIAL NO. | SHIFT / BAY CODE | OPERATOR STAMP | QA INSPECTOR STAMP | DISPOSITION |
|---|---|---|---|---|
| **HVF-2026-586** | SHIFT-A / TI-204 | OP-398 (Verified) | DGQA-48 (Cleared) | **ACCEPTED — PASS** |

**STAGE-GATE WARNING (SECTION 4.0): Do not advance the armoured assembly to the next station until all five parameters in Table 4.0 are counter-stamped in the hull e-Passport ledger.**

# SECTION 5.0 — ROTARY ELECTRICAL SLIP-RING & ELECTRO-HYDRAULIC TRAVERSE DRIVE

**Operational Execution Protocol:** The central multi-channel rotary electrical/fibre-optic slip-ring unit and the stabilized electro-hydraulic/brushless-DC traverse actuator are coupled and verified for signal continuity, high-current power transfer, and deadband response. During shop-floor execution of *Rotary Electrical Slip-Ring & Electro-Hydraulic Traverse Drive*, the designated chargehand shall verify that calibration certificate **CAL-HVF-5683** and material traceability record **LOT-GW-8849** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, voluptas possimus expedita ullam doloribus magni alias numquam deserunt cumque inventore neque ipsam fugiat libero, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

**Quality Assurance & Non-Conformance Control:** Any deviation exceeding the specified tolerance band in Table Rotary must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, explicabo quia sit autem dolor laudantium hic voluptatum culpa maiores fugiat, while automated telemetry from station **CAL-HVF-1795** is archived against hull serial dossier **LOT-NZ-8111** prior to releasing the assembly to the subsequent production workstation.

## 5.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **5.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Voluptatibus numquam impedit eum tempore atque architecto exercitationem corrupti occaecati aut autem repellendus soluta mollitia laborum nisi. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

### TABLE 5.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

| STEP | TECHNICAL PARAMETER / CHECKPOINT | ACCEPTANCE TOLERANCE | METROLOGY / GAUGE | SAMPLING / HOLD |
|---|---|---|---|---|
| **5.1** | Slip-Ring Concentricity | **≤ 0.10 mm to rotation axis** | Dial Indicator on Center Post | Pre-Coupling Check |
| **5.2** | Dynamic Contact Resistance | **≤ 15 mΩ variation at 30 deg/s** | 4-Wire Micro-Ohmmeter | All Power/Signal Rings |
| **5.3** | Insulation Resistance | **≥ 500 MΩ at 500 VDC megger** | Digital Insulation Tester | Channel-to-Chassis |
| **5.4** | Max Slew Rate Velocity | **≥ 40.0 deg/sec stabilized slew** | Gyro Rate Table Telemetry | CW and CCW Full Slew |
| **5.5** | Minimum Tracking Speed | **≤ 0.05 mrad/sec smooth creep** | Autocollimator Target Track | Fire-Control Closed Loop |

| HULL / TURRET SERIAL NO. | SHIFT / BAY CODE | OPERATOR STAMP | QA INSPECTOR STAMP | DISPOSITION |
|---|---|---|---|---|
| **HVF-2026-115** | SHIFT-A / TI-204 | OP-327 (Verified) | DGQA-71 (Cleared) | **ACCEPTED — PASS** |

**STAGE-GATE WARNING (SECTION 5.0): Do not advance the armoured assembly to the next station until all five parameters in Table 5.0 are counter-stamped in the hull e-Passport ledger.**


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# SECTION 6.0 — 360-DEGREE DYNAMIC ROTATIONAL TORQUE & STABILIZATION ACCEPTANCE

**Operational Execution Protocol:** Final turret integration acceptance requires continuous 360-degree rotational torque profiling under simulated hull cant angles (up to 15 degrees) and main gun 2-axis stabilization lock verification on the HVF dynamic tilt test rig. During shop-floor execution of *360-Degree Dynamic Rotational Torque & Stabilization Acceptance*, the designated chargehand shall verify that calibration certificate **CAL-HVF-9943** and material traceability record **LOT-UB-1994** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, recusandae incidunt veniam at placeat in voluptatibus ipsam aperiam labore quia voluptatum illum incidunt architecto, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

**Quality Assurance & Non-Conformance Control:** Any deviation exceeding the specified tolerance band in Table 360-Degree must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, sit doloribus ullam nobis excepturi et ullam odit, while automated telemetry from station **CAL-HVF-6156** is archived against hull serial dossier **LOT-DM-5669** prior to releasing the assembly to the subsequent production workstation.

## 6.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **6.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Nesciunt necessitatibus facere quis harum expedita dolorum laboriosam facere iure nobis sint eos voluptate recusandae blanditiis. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

### TABLE 6.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

| STEP | TECHNICAL PARAMETER / CHECKPOINT | ACCEPTANCE TOLERANCE | METROLOGY / GAUGE | SAMPLING / HOLD |
|---|---|---|---|---|
| **6.1** | Breakaway Torque (Flat Hull) | ≤ **420 Nm CW and CCW** | Inline Rotary Torque Sensor | 8 Azimuth Headings |
| **6.2** | Running Torque Ripple (15° Tilt) | ≤ **± 8% torque fluctuation** | Servo Current Telemetry | 3 Continuous Revolutions |
| **6.3** | Emergency Manual Hand-Crank | ≤ **35 N handle force** | Digital Push-Pull Dynamometer | Full 360° Manual Sweep |
| **6.4** | Azimuth Resolver Accuracy | ≤ **0.15 mrad angular error** | Theodolite Reference Prism | Every 30° Index |
| **6.5** | Turret Travel Lock Engagement | **Zero play; micro-switch interlock** | Mechanical & FCS Interlock | 5 Consecutive Cycles |

| HULL / TURRET SERIAL NO. | SHIFT / BAY CODE | OPERATOR STAMP | QA INSPECTOR STAMP | DISPOSITION |
|---|---|---|---|---|
| **HVF-2026-564** | SHIFT-A / TI-204 | OP-644 (Verified) | DGQA-03 (Cleared) | **ACCEPTED — PASS** |

**STAGE-GATE WARNING (SECTION 6.0): Do not advance the armoured assembly to the next station until all five parameters in Table 6.0 are counter-stamped in the hull e-Passport ledger.**

## APPENDIX A — NORMATIVE DEFENCE STANDARDS, CALIBRATION & REVISION LEDGER

This Appendix establishes the normative statutory, metallurgical, and metrological references governing **Turret Mount System Alignment & Ring Gear Fitting** (HVF/AVNL/SOP-MBT/TI-204) at Heavy Vehicles Factory (HVF), Avadi. All instrumentation utilized in Sections 1.0 through 6.0 must maintain unbroken calibration traceability to National Physical Laboratory (NPL) / NABL standards. Animi dolorem amet facilis sit nulla itaque perspiciatis modi odio amet.

| STD / CODE | STANDARD TITLE & DEFENCE MANUFACTURING APPLICABILITY | CUSTODIAN |
|---|---|---|
| **DGAQA / DGQA-ARM-09** | Quality Assurance Norms for Combat Vehicle Hull, Turret & Running Gear | DGQA (MoD) |
| **MIL-DTL-12560 / DMR** | Armor Plate, Steel, Wrought, Homogeneous & High-Hardness Ballistic Spec | DMRL / HVF Met Lab |
| **ISO 10012 / NABL-141** | Measurement Management Systems — Metrological Confirmation & Calibration | HVF Metrology Cell |
| **AVNL-HSE-SOP-014** | Industrial Safety, Crane Rigging, Fume Extraction & Confined Space Entry | HVF Safety Dept |

## APPENDIX B — MULTI-TIER OFFICIAL SIGN-OFF & AUTHORIZATION MATRIX

This Standard Operating Procedure is officially released for defence production upon execution of the four-tier authorization matrix below by designated HVF, AVNL, and DGQA authorities at Avadi, Chennai.

| TIER / ROLE | DESIGNATION & DIRECTORATE | OFFICER NAME & ID | DIGITAL STAMP / SIGNATURE | DATE |
|---|---|---|---|---|
| **TIER 1: PREPARED BY** | Deputy General Manager (Production Engg) HVF Avadi | Er. R. Venkataraman IOFS-4418 | **[DIGITALLY SIGNED] DGM/PE/HVF-AVADI** | 30-Oct-2026 |
| **TIER 2: REVIEWED BY** | Joint General Manager (Quality & Metallurgy) HVF Avadi | Dr. S. Muthukumar IOFS-3802 | **[DIGITALLY SIGNED] JGM/QA-MET/HVF** | 30-Oct-2026 |
| **TIER 3: CONCURRENCE** | Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles) | Brig. A. K. Rathore CQA(HV) Avadi | **[SEAL & COUNTERSIGNED] CQA(HV)/MOD/GOI** | 30-Oct-2026 |
| **TIER 4: APPROVED BY** | Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL) | Shri K. Parthasarathy CGM / HVF Avadi | **[APPROVED FOR RELEASE] CGM-OFFICE/AVNL** | 30-Oct-2026 |

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